T/O&E DATA BASE DEFINITIONS

"Unit Name" (Field 1): Listed under Field 1 in abbreviated format is the designation of the military unit for which T/O&E data is recorded. This designation matches (without specific numeric identification) the designation entered under the "Unit Name" fields of the ACSDB Unit, Unit Inventory, and Unit Location Data Bases. For example, "AD" is the designation for US and British armored division. In Unit, Unit Inventory, and Unit Location Data Bases are entries for numerous armored divisions, such as "2d AD", "9th AD", and "Guards AD" (2d Armored Division, 9th Armored Division, and Guards Armoured Division). These designations also match entries under the "ATTACHMENTS & DETACHMENTS -- TYPE" field in the Unit Data Base, where the daily order of battles of the US, British, and German armies are described.

Designations used for "Unit Name" were formulated to reflect in abbreviated format the full designation of the unit. To facilitate interpretation of these designations, Attachment 1 lists all designations and their definitions. The full designation for each unit in the T/O&E Data Base, is also presented under "Remarks" in the T/O&E Data Base record.

"Nationality" (Field 2): The nationality of the unit for which T/O&E data is recorded. "A" designates American (US), "B" designates British, and "G" designates German.

"AUTHORIZED EQUIPMENT -- TYPE and AUTHORIZED EQUIPMENT -- AMOUNT" (Fields 3-50): The authorized numbers of major items of equipment, including machine guns, armored vehicles, transport, artillery, etc., for a unit are recorded under Fields 3-50. Under "TYPE" is recorded the system designation, and under "AMOUNT" its authorized number. System designations match identically the designations employed in the ACSDB Weapons and Unit Inventory Data Bases.

"AUTHORIZED PERSONNEL --Total, --Combat Vehicle Crewmen (Armor), --Infantry, --Artillery, --Maintenance, --Maintenance-2, --Medical, --Transportation, --Supply, --Engineer, --Other" (Fields 52-60): The authorized number of personnel of a unit are recorded under these fields. For all units, "AUTHORIZED PERSONNEL --Total" indicates the total personnel strength of a unit. The remaining fields provide the number of personnel of a specific military occupational specialty, as categorized in the ACSDB Unit Data Base narrative. As defined in that narrative, all personnel in independent artillery formations are classified as "Artillery", all personnel in independent tank and tank destroyer formations are "Combat Vehicle Crewmen", etc. For divisions and independent armored and tank brigades only,

personnel strengths are divided into the military occupational specialties.

"AUTHORIZED SUPPLY -- AMMO ON HAND BY TYPE -- Tank/AT" (Field 61a): This is the total weight, in hundreds of pounds, of ammunition carried by a formation for tank guns, antitank guns, man-portable antitank weapons, and certain large caliber antiaircraft guns such as the German 88mm and the US 90mm antiaircraft guns. Details on the categorization of ammunition types for the weapons of various nations are provided in the second, third, and fourth sections of this paper.

"AUTHORIZED SUPPLY -- AMMO ON HAND BY TYPE -- Artillery" (Field 61b): This is the total weight, in hundreds of pounds, of ammunition carried by a formation for artillery guns and howitzers and certain infantry support weapons such as the US 105mm cannon company howitzer (M-3 105mm How) and the German self-propelled 150mm "Brummbaer" infantry support gun (BRUM). See below for additional information.

"AUTHORIZED SUPPLY -- AMMO ON HAND BY TYPE -- Special" (Field 61c): This is the total weight, in hundreds of pounds, of ammunition carried by a formation for rocket projectors. Only the German Army was equipped with rocket projectors (nebelwerfer) in the Ardennes Campaign.

"AUTHORIZED SUPPLY -- AMMO ON HAND BY TYPE -- Other Ammo" (Field 61d): This is the total weight, in hundreds of pounds, of all ammunition other than that in the above three categories carried by a formation. This includes small arms ammunition, demolition explosives, land mines, etc. See below for additional information.

"AUTHORIZED SUPPLY -- Other Supply" (Field 61): This is the total weight, in hundreds of pounds, of all supplies other than ammunition, carried by a formation. These include rations, fodder for German formations using horse-drawn transport, construction supplies, medical supplies, etc. See below for additional information

"AUTHORIZED SUPPLY -- Fuel" (Field 62): This is the total supply, in thousands of gallons, of motor fuel carried by a formation. For US and german units, this includes motor fuel only. For British units it is the estimated amount of all POL (petroleum, oil, and lubricants).

"Remarks" (Field 63): Entered under "Remarks" is information in text format which contains the full designation of the unit and qualifies or amplifies data in any other fields.

"Sources" (Field 64): Entries under "Sources" match exactly entries under "Source Number" in the ACSDB Reference Data

Base, where information on sources used for T/O&E data is recorded. A prefix of "TU" is assigned to US T/O&Es, "TB" to British T/O&Es, and "TG" to German T/O&Es. Each prefix is followed by a unique two-digit number (for US T/O&Es) or three-digit number (for German T/O&Es). Note that all entries under "Sources" for British T/O&Es are "TB500". This is because individual Reference Data Base records are not written for the British T/O&Es; their sources and compilation are listed and explained entirely in this paper.

ADDITIONAL INFORMATION ON BRITISH T/O&Es

Data on British T/O&Es from battalion- up to division-level is provided in the T/O&E Data Base for infantry formations. Data is also provided for T/O&Es of other key combat units (primarily armored units) and service and support units.

British T/O&Es are not referenced by individual ACSDB Reference Data Base records. (Entered under "Sources" for all British T/O&Es is the entry "TB500".) This discussion serves as an explanation of how the British T/O&Es are compiled, any estimation methodologies used in their compilation, and the sources used in their derivation.

The sources used in compilation of the British T/O&Es are:

- Ellis, Chris and Peter Chamberlain (eds.). Handbook on the British Army, 1943. London: Arms and Armour Press, 1975. (This is a reprint of the US Army's TM 30-410, Handbook on the British Army, originally produced in 1943.) Hereafter cited as Handbook.

- Ellis, Major L.F. The Battle of Normandy. Vol. I. Victory in the West, History of the Second World War, UK Military Series. London: Her Majesty's Stationery Office, 1962. Hereafter cited as Ellis.

- Joslen, Lieut-Colonel H.F. Orders of Battle: United Kingdom and Colonial Formations and Units in the Second World War, 1939-1945. Vols. I & II. London: Her Majesty's Stationery Office, 1960. Hereafter cited as Joslen.

- Miscellaneous Records of the British 5th Infantry Division in the Italian Campaign, 1943-1944. (These records were obtained from the British Public Records Office during the 1960s and are kept in File Folder No. 181 at the Historical Evaluation and Research Organization's offices. Their exact archival identification is not known. They include "Strengths of Main Bodies 5 Div Less 15 Ind Bde Gp Adv Parties and Rear Parties - Appx 'A' to 5 Div Movement Control Order No. 9 dated 24 Jun 44", and "Composition of 15 Inf Bde Gp - Appendix 'A' to 5 Div Movement Control Order No. 1 dated 15 June 1944.") Hereafter cited as 5th ID.

Joslen is the major source of information on vehicle and total personnel strengths for the British infantry, armoured, and airborne division T/O&Es (ID, AD, and AbnD). Specific MOS strengths for the divisions are estimated from the strengths of the divisions' component units. In some cases, strengths of comparable US units are used for estimation. For example, the

personnel and equipment strengths of the field hygiene section of the British infantry division is estimated to be one-fourth of that of the US sanitary company, a unit which had a similar function. Other MOS data is estimated from data on British units with similar functions. For example, the airborne division's parachute engineer squadron is estimated as three-fourths of an engineer field squadron of a standard British infantry battalion. (A parachute battalion had approximately three-fourths the personnel strength of an infantry battalion.)

The major source of data for the infantry battalion, motor infantry battalion, reconnaissance battalion, artillery, and engineer units -- InfBn, MotorInfBn, RcnBn, FldReg (Royal Arty), MedReg (Royal Arty), HvyReg (Royal Arty), FPkCoRE, FldCoRE -- is Handbook. For the machine gun battalion (MGBn), Handbook provides data on a battalion with four machine gun companies. The 1944 organization of this unit was three MG companies and one 4.2" mortar company. Appendix 4 of Ellis contains the strengths of these company-size units.

For the antiaircraft regiments, the antitank regiment (HAAReg, LAAReg, and ATReg), and divisional medical, maintenance and supply units, 5th ID is the main source of data. Data on transport for the AT Reg, as well as the MedReg and HvyReg (Royal Arty), is estimated using the transport strength of the FldReg (Royal Arty) from Handbook, pgs. 30-33.

For British non-divisional medical units (FldAmbl, LtFldAMbl, and FldDsSt), data from T/O&Es of similar US units is used for estimation. For example, the field ambulance (FldAmbl) data is estimated from that of the US clearing and collecting companies, the functions of which the British unit combined. (Ambulance strength for this unit is from Handbook).

Handbook provides data on the T/O&Es of the independent armoured brigade, the infantry brigade, and the tank brigade -- ArmBde (Independent), InfBde, and TkBde. Total softskin transport vehicle strength of the ArmBde (Independent) is given as approximately 1,200 in Ellis. The distribution of vehicle types is estimated by subtracting the number of identified vehicle types of the motor battalion and the number of armored fighting vehicles from the total strength (1,200 - 483 = 717). A ratio for determining numbers of specific vehicle types is established by dividing 717 into the total number of softskin transport vehicles in a British armoured division (2,601) -- 0.28. This ratio was then applied to the numbers of specific softskin transport vehicle types to calculate numbers of these vehicle types in the brigade. A similar approach is used to estimate the number of vehicles in the tank brigade T/O&E.

Personnel data for the armoured car regiment (ACReg) is taken from Ellis, while softskin vehicle strengths are estimated

as that of a British infantry division's reconnaissance battalion without the T16s and M-3 HTs.

No Maintenance-2 MOS strengths are estimated for British T/O&Es. As discussed in the ACSDB Unit Data Base narrative, MOS strengths are not estimated for British units in the Unit Data Base. The MOS strengths provided here may be used for estimation purposes with British units in the Unit Data Base.

British T/O&E fuel supply data is estimated by using the POL (petroleum, oil, and lubricants) computations for similar US vehicles found in US Army FM 101-10, 21 December 1944, Chapter 3, pg. 12. Handbook, pg. 84, gives the following principles of supply to be followed for gasoline in units: "...all vehicles are filled and all reserves are made complete...adequate reserve is maintained on wheels in front of railhead (sufficient for 75 miles, carried by petrol companies)." The fuel amounts carried in the fuel tanks of each similar vehicle type are multiplied by the number of vehicles in the British T/O&E. To this figure is added the fuel amounts required to move the total numbers of vehicles 75 miles, as presented in columns 4-7 of FM 101-10, Chapter 3, pg. 12. The total figure is the estimated authorized POL level of the unit.

British T/O&E ammunition supply data is estimated from the basic load data (first and second echelon) for British weapons as recorded in the ACSDB Weapons Data Base. Upon comparison with ammunition supply levels of comparable US units, it was determined that the amounts in first and second echelon are probably one-half of the total unit load. Therefore, the amounts calculated for first and second echelon are multiplied by two to give the estimated total unit ammunition levels.

British terminology classifies tanks into an "Infantry" and a "Cruiser" category. These terms are used in the T/O&E data base for British units in the following manner:

- British AD.
Cruiser Tank: M-4 75mm; Sherman 17-pr (M-4A4); Cromwell 75mm (A27); Cromwell 95mm (A27); A-30 Challenger

- British AbnD.
Cruiser Tank: M-4 75mm; Cromwell 75mm (A27)

- TkBde.
Infantry Tank: Churchill 6-pr (A22); Churchill 75mm (A22); Churchill 95mm (A22)
Cruiser Tank: M-4 75mm; Sherman 17-pr (M-4A4)

- ArmBde (Independent). See under "Remarks" in the T/O&E Data Base.

For calculating British T/O&E ammunition supply levels by type ("Tank/AT", "Arty", and "Other"), the following categorization is used:

Tank/AT
Tank gun ammunition
Self-propelled antitank gun ammunition
Towed antitank gun ammunition
PIAT ammunition
Armored car armament ammunition greater than .50 caliber

Arty
Howitzer and gun (SP and towed) ammunition
Mtr 4.2" ammunition

Other
Machine gun ammunition
Armored/scout car machine gun ammunition (incl. .50 cal. MG)
Antiaircraft artillery ammunition, including the 3.7" AA (T)

This same categorization scheme is used for British unit ammunition data recorded in the ACSDB Unit Data Base. Note that the ammunition for the British 3.7" AA (T) (the 3.7" antiaircraft gun) is counted as "Other" ammunition, unlike the ammunition of its US and German equivalents, the 90mm and 88mm antiaircraft gun, which is considered antitank ammunition in the ACSDB. Although the British piece had the capability of engaging armor, British doctrine called for its use in its primary antiaircraft role and only in emergencies in an antitank role. British XXX Corps ammunition expenditure records show a relatively minor expenditure of 3.7" ammunition during the Ardennes Campaign period, all after the German armor threat had dissipated. No mention was made in British operations orders of using 3.7" guns in the antitank role, unlike US orders for the 90mm piece. The German 88mm gun's employment as an antitank weapon is well documented.

No authorized weights are provided for the "Other Supply" category of British T/O&Es. This is because no information was found which provided exact data or could be used to estimate the data. In the ACSDB Unit Data Base, "Other Supply" data is also not provided for British units. If necessary, the estimation process used for US units could be applied to estimate British "Other Supply" data, i.e., total personnel x 24.99 pounds per man. See the above discussion on US T/O&Es for additional information.

Attachment 1

List of T/O&E Unit Designations/Abbreviations

American

AAABdeH	Antiaircraft Artillery Brigade Headquarters
AAABNAW	Antiaircraft Artillery Battalion, Automatic Weapons
AAABN G	Antiaircraft Artillery Battalion, Gun
AAABNSP	Self-propelled Antiaircraft Artillery Battalion
AAAGrpHQ	Antiaircraft Artillery Group Headquarters
AbnD	Airborne Division
AD (Heavy)	Heavy Armored Division
AD (Standard)	Standard Armored Division
ArmFABN	Armored Field Artillery Battalion
ArmGrpH	Armored Group Headquarters
ArmInfB	Armored Infantry Battalion
CavGrpH	Cavalry Group Headquarters
CavSq	Cavalry Squadron
CavTrp	Cavalry Troop
ClrngCo	Medical Clearing Company
CollCo	Medical Collecting Company
EngBN	Combat Engineer Battalion
EngBNC	Construction Engineer Battalion
EngCo	Combat Engineer Company
EngGrpH	Engineer Group Headquarters
EngReg	Engineer General Service Regiment
EvacHosp	Evacuation Hospital
FABde	Field Artillery Brigade Headquarters
FABN105	105mm Howitzer Field Artillery Battalion
FABN155	155mm Gun or Howitzer Field Artillery Battalion
FABN240	240mm Howitzer Field Artillery Battalion
FABN4.5	4.5" Gun Field Artillery Battalion
FABN8"G	8" Gun Field Artillery Battalion
FABN8"H	8" Howitzer Field Artillery Battalion
FABNSP	Self-propelled 155mm Gun Field Artillery Battalion
FAGrpHQ	Field Artillery Group Headquarters
FAObsBN	Field Artillery Observation Battalion
FAObsBt	Field Artillery Observation Battery
FldHosp	Field Hospital
GIReg	Glider Infantry Regiment
HAMCo	Heavy Automotive Maintenance Company
HMCo (FA)	Heavy Maintenance Company (Field Army)
HMCo (Tk)	Heavy Maintenance Company (Tank)
ID	Infantry Division
InfBN	Infantry Battalion

InfReg	Infantry Regiment
MaintCo (AA)	Maintenance Company (Antiaircraft)
MAMCo	Medium Automotive Maintenance Company
MMaintCo	Medium Maintenance Company
MrtBN	Chemical Mortar Battalion
MrtCo	Chemical Mortar Company
PFABN	Parachute Field Artillery Battalion
PIBN	Parachute Infantry Battalion
PIReg	Parachute Infantry Regiment
QMTrkCo	Quartermaster Truck Company
QMTrlBN	Quartermaster Trailer Battalion
RcnTrp	Reconnaissance Troop
RgrBN	Ranger Battalion
TDBNSP	Self-propelled Tank Destroyer Battalion
TDBNT	Towed Tank Destroyer Battalion
TDCoSP	Self-propelled Tank Destroyer Company
TDCoT	Towed Tank Destroyer Company
TDGrpHQ	Tank Destroyer Group Headquarters
TkBN	Tank Battalion
TkBN (Lt)	Light Tank Battalion
TkBNMX	Mine-exploding Tank Battalion
TkCo	Tank Company
TkCoMX	Mine-exploding Tank Battalion

The following suffixes are used with US field and antiaircraft artillery units: (Gun-Tractor); (Gun-Truck); (HOW-Tractor); (HOW-Truck); (Mbl); (SM); (Tractor); and (Truck). Gun tractor-drawn; Gun truck-drawn; Howitzer tractor-drawn; Howitzer truck-drawn; Mobile; Semimobile; Tractor-drawn; and Truck-drawn.

British

AbnD	Airborne Division
ACReg	Armoured Car Regiment
AD	Armoured Division
ArmBde (Independent)	Independent Armoured Brigade
ATReg	Antitank Regiment
FldAmbl	Field Ambulance
FldCoRE	Field Company, Royal Engineers
FldDsSt	Field Dressing Station
FldReg (Royal Artillery)	Field Regiment, Royal Artillery
FPkCoRE	Field Park Company, Royal Engineers
HAAReg	Heavy Antiaircraft Regiment
HvyReg (Royal Artillery)	Heavy Regiment, Royal Artillery
ID	Infantry Division
InfBde	Infantry Brigade
InfBn	Infantry Battalion
LAAReg	Light Antiaircraft Regiment
LtFldAmbl	Light Field Ambulance
MedReg (Royal Artillery)	Medium Regiment, Royal Artillery

MGBn	Machine Gun Battalion
MotorInfBn	Motor Infantry Battalion
MtzdInfBn	Motorized Infantry Battalion
RcnBn (RcnReg)	Reconnaissance Battalion (Regiment)
TkBde	Tank Brigade

German

AArtyBN	Army Artillery Battalion
AArtyBt	Army Artillery Battery
AMrtBN	Army Mortar Battalion
ASvcS	Army Service School
BrB(GE)	Light bridge equipment column (company) non-motorized
BrB(m)	Light bridge column (company) fully motorized
BrJ(GE)	Heavy bridge equipment column (company) non-motorized
BrJ(m)	Heavy bridge column (company) fully motorized
EngBde	Engineer Brigade
EngBN	Combat Engineer Battalion
EngBN C	Construction Engineer Battalion
EngBrBN	Engineer Bridge Battalion
EngEABN	Engineer Ersatz und Ausbildung (Replacement and Training) Battalion
FJArtR	Fallschirmjaeger (Paratrooper) Artillery Regiment
FJBN	Fallschirmjaeger (Paratrooper) Battalion
FJD	Fallschirmjaeger (Paratrooper) Division
FJReg	Fallschirmjaeger (Paratrooper) Regiment
FLAKBde	Fliegerabwehrkanone (Antiaircraft) Brigade
FLAKBN	Fliegerabwehrkanone (Antiaircraft) Battalion
FLAKCo	Fliegerabwehrkanone (Antiaircraft) Company
FLAKReg	Fliegerabwehrkanone (Antiaircraft) Regiment
FLAKStR	Fliegerabwehrkanone Sturm (Antiaircraft Assault) Regiment
FstArBN	Festung (Fortress) Artillery Battalion
FstArBt	Festung (Fortress) Artillery Battery
FstMGBN	Festung (Fortress) Machine Gun Battalion
FstPAKC	Festung Panzerabwehrkanone (Fortress Antitank) Company
FstPnBN	Festung (Fortress) Penal Battalion
HyPzBN	Heavy Panzer (Armored) Battalion
HyPjBN	Heavy Panzerjaeger (Antitank) Battalion
ID	Infantry Division
OTBde	Organization Todt (paramilitary civilian construction corps) Brigade

OTBN	Organization Todt (paramilitary civilian construction corps) Battalion
OTReg	Organization Todt (paramilitary civilian construction corps) Regiment
PjBN	Self-propelled Panzerjaeger (Antitank) Battalion
PjBNT	Towed Panzerjaeger (Antitank) Battalion
PjCo	Self-propelled Panzerjaeger (Antitank) Company
PzCo	Panzer (Armored) Company
PzEngCo	Panzer (Armored) Engineer Company
PzGD	Panzer Grenadier (Armored Infantry) Division
PzD	Panzer (Armored) Division
SSArtBN	SS Artillery Battalion
SSPzD	SS Panzer (Armored) Division
SSWfrBN	SS Werfer (Rocket Launcher) Battalion
StgBde	Sturmgeschuetz (Assault Gun) Brigade
StMrsCo	Sturm Moerser (Assault Mortar) Company
StPzBN	Sturm Panzer (Assault Vehicle) Battalion
VAK	Volksartillerie Korps ("People's" Artillery Corps)
VABN	Volksartillerie ("People's" Artillery) Battalion
VGD	Volks Grenadier ("People's" Infantry) Division
VWB	Volkswerfer ("People's" Rocket Launcher) Brigade
VWBN	Volkswerfer ("People's" Rocket Launcher) Battalion
VWBty	Volkswerfer ("People's" Rocket Launcher) Battery
VTB	Volga Tartar Battalion

Note: SS = Waffen SS (Schutz Staffeln) field formations of the SS.

The ACSDB Weapons Data Base

Major items of equipment (weapons, vehicles, and aircraft) are identified and recorded in the T/O&E, Unit Inventory, and Air Data Bases of the ACSDB. These three data bases focus on the numerical strengths and losses of the items of equipment (including damage and in-repair status of selected systems). Performance and descriptive data on the items of equipment themselves is recorded in the Weapons Data Base.

There are 209 systems in the Weapons Data Base. These include all primary weapons, vehicles, and aircraft of the US, British, and German formations which operated in the Ardennes. Each system is identified by a unique designation, recorded under "Equipment Name" (Field 1) in the Weapons Data Base. In all but one case, there is one record per system in the Weapons Data Base. The one exception is the German cargo carrier truck, identified as "LKW" in the ACSDB (for lastkraftwagen, or cargo truck). German documents and records do not distinguish between the various kinds of cargo trucks used by German units. Data is therefore provided for two of the most common variants in use.

The unique designations that identify the systems in the Weapons Data Base match exactly the designations recorded in the T/O&E and Weapons Data Bases of the ACSDB. For example, "M-4 75mm", the designation assigned to the M-4 "Sherman" medium tank used by US and British forces, is found in the Weapons Data Base, in Unit Inventory Data Base records of units equipped with this tank, and in T/O&E Data Base records of armored divisions and tank battalions equipped with it. The designations assigned to the systems were created to reflect in abbreviated format the full designation of the system, and to fit into the limited space available in the fields of the various data bases. Attachment 1 is a complete list of all weapon, vehicle, and aircraft designations in use in the ACSDB.

Under "Weapon System Description" and "Remarks" (Fields 18 and 30) is recorded information which describes the system. For some US and British systems additional text is provided in Attachment 2. This text was added because limited space in Fields 18 and 30 in some cases did not permit the inclusion of complete information on the system. Therefore, it is critical to a comprehensive review and to the most efficient use of the ACSDB Weapons Data Base that this paper be consulted. Additional descriptive information may also be found for some systems under "Remarks" in the Reference Data Base. (The data for each Weapons Data Base record is referenced by one record in the Reference Data Base.)

The Single-Shot Probability of Kill (SSPK) narrative prepared for the ACSDB should also be consulted when reviewing and using the Weapons Data Base, as should documents referenced as sources of data in that paper and for the Weapons Data Base. Much information is contained in these sources which, due to the format limitations of the Weapons Data Base records, could not be included in the Weapons Data Base. The subject of World War II-era weapons is a complex one, and all pertinent sources of information should be examined when studying the it.

Provided below are definitions of the fields used in the Weapons Data Base to record data on systems characteristics and specifications.

"Equipment Name" (Field 1): The unique designation assigned to each item of equipment recorded in the ACSDB. Attachment 1 is a complete listing of the equipment designations. Data was not recorded in the Weapons Data Base for most system variants or submodels, because they are not identified in the historical records of units. For example, there were three different production submodels of the German Panzerkampfwagen V "Panther" medium tank, the D, the A, and the G models. These differed in minor details such as secondary armament and commander's cupola design. The records of German units do not identify the submodel of the tank, but rather simply identify the tank as "PzKw V" or "V". Therefore, data for all three submodels of the Panther tank is not recorded in the Weapons Data Base.

"Nationality" (No field number): The nationality of the military force which employed the system in the Ardennes Campaign. "A" designates American (US), "B" designates British, and "G" designates German. Some systems were used by both US and British forces, such as the ".25 ton 4x4" (the 1/4 ton jeep) and the "M-4 75mm" (the M-4 Sherman medium tank). In these cases, the nationality of the manufacturing nation is used.

"Fuel Consumption Factor" (Field 2): This data is provided in either gallons used per 12 hours (gallons/12 hours) or miles per gallon (miles/gal). These values represent fuel consumption at cruising speed for aircraft and under average road conditions at moderate speeds for vehicles, except where otherwise specified under "Remarks".

"Ammo Weight Carried" (Field 3): Ammunition weight carried is provided in hundreds of pounds for systems which possess organic armament. This is the weight of the basic ammunition load of the system. For systems with more than one armament, it is the weight of the basic load of the primary armament, i.e., bomb load for aircraft and main gun ammunition for tanks. This is a computed value arrived at by multiplying the number of rounds carried by the weight of the individual round.

"Weapon Weight" (Field 4): This is the total weight in pounds of the entire system. Unless otherwise indicated under "Remarks", this weight is for systems with full fuel and ammunition loads.

"Individual Round Weight" (Field 5): This is the weight in pounds of the primary ammunition round used by systems which carry armament. This is complete round weight, including propellant, projectile, and casing. For aircraft, this is the individual weight of the most common bomb carried by the system. In the Weapons Data Base, the primary armament of aircraft is the bomb. Data on machine guns or cannons mounted on aircraft is recorded as secondary armament.

"Basic Load" (Field 6): Basic load is the number of rounds of ammunition either carried by the system, or carried by the system and carried in the trains of the unit equipped with the system. For combat vehicles, such as tanks, tank destroyers, and self-propelled artillery, the data provided is actually the ammunition load stowed on the vehicle, a figure normally significantly less than the basic load. For towed artillery and antitank guns, particularly in the case of the Germans, the information provided under "Basic Load" is of two different kinds. This is due to the fact that different sources were utilized, out of necessity, to generate basic load data. For data referenced by the source RH3/v.135, Munitions Allotment Tables for German Divisions, the figure provided is for number of rounds with the man or on the gun, i.e., weapon ("Mann, Gesch."). On systems that have the basic load field referenced by US National Archives Microfilm # T314,1594,1166 or T314,1134,232, or by "Ger. Army Equipment" (any of the volumes of the source Illustrated Record of German Army Equipment), the information provided reflects total basic load for the system, and not merely what was with the man or weapon. For many of the US systems, a breakdown between total basic load and ammunition forward with the man or weapon does exist. In these cases, the ammunition amount forward with the system is provided under "Basic Load" and the complete basic load is provided under "Remarks". In all cases, the "Remarks" field and any source(s) used for data should be consulted for clarification. For aircraft, data provided under "Basic Load" is the total number of bombs carried by the aircraft. If this data is maximum bomb load, it is so indicated under "Remarks". Alternate bomb load configurations, when known, are provided under "Remarks".

"Maximum Effective Range" (Field 7): In the sources used for weapons data, a variety of information is provided for ranges. These include vertical range for antiaircraft systems, maximum range for artillery pieces, and effective range against various targets for tank and antitank guns. The value entered under "Maximum Effective Range" is described under "Remarks" if it means anything other than the maximum effective range. See the SSPK narrative for further discussion on this subject.

"Failure on Issue (RAM)" (Field 8): This value represents the frequency of weapons failures due to failures intrinsic to the weapon itself. In no cases in the Weapons Data Base does this value represent reliability of drive train or automotive components; it pertains only to armament. For aircraft bombs, data is in some cases provided for the reliability of bomb release mechanisms, as indicated under "Remarks". In fact, only limited data was found on reliability of systems, and the SSPK narrative should be consulted for additional information on the subject.

"Crew Size" (Field 9): Crew size is the number of personnel required to operate the system as intended by design.

"Crew Specialty" (Field 10): Entries in this field indicate the so-called military occupational specialty (MOS) designations employed in the ACSDB. For definitions of these designations see the ACSDB Unit Data Base narrative. Entries under this field are made only for systems employed exclusively by units of a single MOS type, such as artillery pieces. Many systems were employed by artillery, infantry, armored, engineer, and other units, i.e., the ".25 ton 4x4" (the jeep) in widespread use with the US and British armies. The following abbreviations are entered under "Crew Specialty":

- Airmen = Airmen (Aircraft Crewmen)
- Arty = Artillery
- CVC = Combat Vehicle Crewmen (Armor)
- Eng = Engineer
- Inf = Infantry
- Maint = Maintenance
- Other = Other, primarily Antiaircraft Artillery

"Sustained Rate of Fire" (Field 11): In most cases, the practical rate of fire which a weapon can fire in rounds per minute is entered under "Sustained Rate of Fire". For some systems, only the cyclic rate of fire is available in data sources. In these cases, the type of rate of fire is described under "Remarks". For aircraft, sustained rate of fire is equal to the standard bomb load carried by the aircraft.

"Sensor Type/Characteristics" (Field 12): Sensor type describes the sighting or detection system attached to or most commonly used in conjunction with a weapon. If available, additional data, such as the magnification power of the sight, is provided in Attachment 3.

"Sensor Range" (Field 13): Sensor range is the maximum distance in meters at which the most commonly engaged target can be detected by the sensor equipment organic to the weapon, under ideal environmental conditions.

"Off-road Movement Rate" (Field 14): Off-road movement rate is the speed in kilometers per hour at which a vehicle can move over cross-country terrain. For aircraft, speed is always recorded under "Off-road Movement Rate". Aircraft speed recorded in the Weapons Data Base is usually maximum speed, although cruising speed is also in some cases provided under "Remarks". Movement rates are provided for some towed systems as the speed of the towing vehicle with the system in tow. Standard movement rates of infantry, taken from World War II-era field manuals, are also provided for certain man-portable systems.

"On-road Movement Rate" (Field 15): On-road movement rate is the speed in kilometers per hour at which a vehicle can move over paved or hard-road surfaces. In general, this is equal to maximum speed in data source.

"Weapon Category" (Field 16): An abbreviated code is provided in this field to identify the type of system. The codes and their definitions are as follow:

- AAA-SP Self-propelled Antiaircraft Artillery
- AAA-T Towed Antiaircraft Artillery
- AC-B Bomber Aircraft
- AC-F Fighter Aircraft
- AC-FB Fighter-bomber Aircraft
- AC-LN Liaison Aircraft
- AC-NF Night-fighter Aircraft
- AC-T Transport Aircraft
- ArmCar Armored Car
- Arty-T Towed Artillery
- AT-Inf Man-portable Antitank Weapon
- AT-SP Self-propelled Antitank Gun
- AT-T Towed Antitank Gun
- Half-T Half-track
- HT Heavy Tank
- Inf-SP Self-propelled Infantry Support Weapon
- LT Light Tank
- Misc Miscellaneous (includes Flamethrowers, Ammunition Carriers, Tracked Personnel Carriers, Command Vehicles, Cranes, Demolition Vehicles, Motorcycles, and Horse-drawn Vehicles)
- MG Machine Gun
- Mortar Mortar
- MT Medium Tank
- PM Prime Mover
- Rkt-SP Self-propelled Rocket Launcher
- Rkt-T Towed Rocket Launcher
- RRArty Railroad Artillery
- RV Recovery Vehicle
- SPArty Self-propelled Artillery
- Trl Trailer
- Truck Wheeled Personnel or Cargo Carrier

"Sources" (Field 17): Entries here match exactly entries under "Source Number" in the Reference Data Base of the ACSDB, where information on sources used for weapons data is recorded. A prefix of "WA" is assigned to American (US) systems, "WB" for British systems, and "WG" for German systems. Each prefix is followed by a unique three-digit number. Note that all of the numbers are not consecutive. This is due to the fact that the ACSDB weapons list was drafted several times before a final, comprehensive list was prepared.

"Remarks" (Field 18): Entered under "Remarks" is information in text format which describes the system and qualifies or amplifies data in any of the other fields. For vehicles and aircraft with secondary armament, the secondary armament is briefly described here.

"Burst Area" (Field 19): The area of burst in square meters is provided in this field for systems which employ high-explosive ammunition, such as artillery and aircraft. For further discussion on this subject, consult the SSPK narrative.

"ANTIAIRCRAFT DATA -- Salvo Size" (Field 20): For magazine-fed, small-caliber antiaircraft guns, the capacity of the magazine, if known, is provided in this field. This data requirement, and the other requirements prefaced by "ANTIAIRCRAFT DATA", may be more relevant to modern surface-to-air missiles (SAMs). Therefore, any data entered under these fields may not have direct application in the ACSDB.

"ANTIAIRCRAFT DATA -- Time Between Reloads" (Field 21): No data is entered under this field, as it could not be found in research.

"ANTIAIRCRAFT DATA -- Shots Before Reloads" (Field 22): For belt- and magazine-fed antiaircraft systems (including machine guns), the belt or magazine capacity in rounds is provided under this field, if the data is available.

"ANTIAIRCRAFT DATA -- Maximum Targets" (Field 23): This is the maximum number of targets that an antiaircraft systems can engage. For most systems recorded in the ACSDB, this value should be one (1). Some large-caliber antiaircraft guns had radar direction capability, but no data was found on the number of systems that they could engage simultaneously.

"AIRCRAFT DATA -- Fuel Requirements" (Field 24): This is the maximum amount of fuel, in pounds, that an aircraft's fuel tank(s) can hold.

"AIRCRAFT DATA -- Range" (Field 25): This is the straight-line distance, in kilometers, that an aircraft can fly on a full load of fuel. It is not equal to combat radius.

"AIRCRAFT DATA -- Runway Length Needed" (Field 26): This is the distance, in feet, that an aircraft requires to take off or land.

"AIRCRAFT DATA -- Dry Cargo Capacity" (Field 27): This is the weight, in tons, of dry supplies that an aircraft can carry. This data is provided for only six aircraft in the ACSDB: C-47 (transport); Ju-52 (transport); B-17 (heavy bomber); B-24 (heavy bomber); Halifax (heavy bomber); and Lancaster (heavy bomber). For the bombers, this field is computed by subtracting the empty weight of the aircraft from the maximum loaded weight of the aircraft.

"AIRCRAFT DATA -- Wet Cargo Capacity" (Field 28): This is the weight, in tons, of wet supplies that an aircraft can carry. No information relevant to this data requirement was found in research for aircraft in the ACSDB.

"AIRCRAFT DATA -- Crew Rest Requirements" (Field 29): This is the maximum time, in days, that an aircrew can fly an aircraft. No data on crew rest requirements was found in research for aircraft in the ACSDB.

"Weapon System Description" (Field 30): This is a brief description of the system and includes, for weapons, the caliber (bore diameter) of the system (for tanks, the caliber of the main gun). A complete list of weapon systems descriptions is provided in Attachment 1.

SECONDARY ARMAMENT DATA #1 and SECONDARY ARMAMENT DATA #2 (Fields 31-42): For combat vehicles and aircraft with more than one type of armament, data is recorded under fields 31-42. This includes data for machine guns on tanks and machine guns and cannons on aircraft. Definitions of these fields are identical to those provided above, with the exception of "Weapon Category" (Field 31), which is defined as an abbreviation identifying the type of armament. The four abbreviations used are:

- B Aerial Bomb
- C Cannon
- Misc Miscellaneous (i.e., Flamethrower)
- MG Machine Gun

Primary and secondary sources were researched for data recorded in the Weapons Data Base. Data was not found for all data requirement for all systems. To reduce this lack of data for British systems, substitutions were made for requirements using suitable data from other German, US, and British systems. Attachment 4 lists the fields filled with substitute data and the systems from which this data was obtained.

Some data was not found for the ACSDB Weapons Data Base and could not be substituted with data from other systems. The

fields for which data is not available are filled with "9s" (i.e., "99", "999", "9999", etc.). Additional research may obtain this data,or further substitutions may be made.

Attachment 1

ACSDB Weapons Designations

US Designations

INFANTRY WEAPONS

M-1919 A4/A5 MMG	US M-1919 A4/A5 .30 caliber machine gun
M-2 .50 HMG	US M-2 .50 caliber heavy machine gun
M-9 2.36" Rkt	US M-9 2.36" rocket launcher ("bazooka")
M-2 60mm Mtr	US M-2 60mm mortar
M-1 81mm Mtr	US M-1 81mm mortar
M-1 4.2" Mtr	US M-1 4.2" chemical mortar

ANTIAIRCRAFT ARTILLERY

M-55 4x.50	US M-55 quad .50 caliber towed heavy machine guns
M-16 4x.50	US M-16 quad .50 caliber self-propelled heavy machine guns
M-1 37mm AAA	US M-1 37mm towed antiaircraft gun
M-15 37mm	US M-15 37mm self-propelled antiaircraft gun w/twin .50 caliber heavy machine guns
M-1 40mm AAA	US M-1 40mm towed antiaircraft gun
M-1A1 90mm AAA	US M-1A1 90mm towed antiaircraft gun

ANTITANK (TOWED)

M-3A1 37mm AT	US M-3A1 37mm towed antitank gun
M-1 57mm AT	US M-1 57mm towed antitank gun
M-5 3" AT	US M-5 3" towed antitank gun

ARTILLERY

M-1 75mm How	US M-1 75mm pack howitzer
M-8 75mm SP	US M-8 75mm self-propelled howitzer
M-2A1 105mm How	US M-2 105mm towed howitzer
M-3 105mm How	US M-3 105mm towed infantry howitzer
M-7 105mm SP	US M-7 105mm self-propelled howitzer ("Priest")
M-1 4.5" Gun	US M-1 4.5" gun
M-1A1 155mm Gun	US M-1A1 155mm towed gun
M-12 155mm SP	US M-12 155mm self-propelled gun
M-1 155mm How	US M-1 155mm towed howitzer

M-1 8" How	US M-1 8" towed howitzer
M-1 8" Gun	US M-1 8" towed gun
M-1 240mm How	US M-1 240mm towed howitzer

ANTITANK AND INFANTRY SUPPORT (SELF-PROPELLED)

M-10	US M-10 3" tank destroyer
M-18	US M-18 76mm tank destroyer ("Hellcat")
M-36	US M-36 90mm tank destroyer ("Jackson")

TANKS

M-5 Tk	US M-5A1 light tank w/37mm gun ("Stuart")
M-24	US M-24 light tank w/75mm gun ("Chaffee")
M-4 75mm	US M-4 medium tank w/75mm gun ("Sherman")
M-4 76mm	US M-4 medium tank w/76mm gun ("Sherman")
M-4 A3E2	US M-4 medium tank w/75mm or 76mm gun ("Jumbo")
M-4 105mm	US M-4 medium tank w/105mm howitzer
M-4 MX	US M-4 medium tank w/mine exploding device
*M-4 Dozer	US M-4 medium tank w/bulldozer blade attached

HALF-TRACKS

M-3 HT	US M-3 half-track
M-21 81mm	US M-21 81mm half-track mounted mortar

ARMORED CARS

M-8 AC	US M-8 armored car w/37mm gun ("Greyhound")
M-20 AC	US M-20 armored utility vehicle

SOFTSKIN VEHICLES

.25 ton 4x4	US 1/4 ton truck ("jeep")
.25 ton trl	US 1/4 ton trailer
.75 ton 4x4	US 3/4 ton truck
*.75 ton Ambl	US 3/4 ton ambulance
1 ton trl	US 1 ton 2-wheeled trailer
1.5 ton 6x6	US 1-1/2 ton truck
2.5 ton 6x6	US 2-1/2 ton truck
2.5 ton Dump	US 2-1/2 ton dump truck

2.5 ton SWB	US 2-1/2 ton truck w/short wheel base
*2.5 ton wrecker	US 2-1/2 ton wrecker
2.5 ton trl	US 2-1/2 ton 4-wheeled trailer
750 gal trk	US 2-1/2 ton truck w/750 gallon tank (gasoline)
*700 gal trk	US truck w/700 gallon tank (water)
250 gal trl	US 250 gallon trailer (water)
4 ton 6x6	US 4 ton truck
4 ton trl	US 4 ton 2-wheeled trailer
4 ton wrecker	US 4 ton wrecker
5 ton 4x2 Tractor	US 5 ton tractor-trailer
*5 ton trl	US 5 ton trailer
6 ton 6x6	US 6 ton truck
7.5 ton 6x6	US 7-1/2 ton truck
8 ton trl	US 8 ton 4-wheeled trailer
10 ton wrecker	US 10 ton wrecker
40 ton tractor trailer	US 40 ton tank transporter/tractor-trailer
M-4 Tractor	US M-4 18 ton tracked prime mover
M-5 Tractor	US M-5 13 ton tracked prime mover
Clamshell trl	US trailer used to carry clamshell bucket used w/crane
crane	US truck-mounted crane
*DUKW	US 2-1/2 ton amphibious truck

MISCELLANEOUS ARMORED VEHICLES

M-30	US M-30 ammunition carrier used w/M-12 155mm SP
M-32	US M-32 tank recovery vehicle w/M-1 81mm Mtr
T-41	US T-41 tracked prime mover/armored utility vehicle

AIRCRAFT

Light Airplane	US Piper L-4 single-engine observation aircraft ("Grasshopper")
A-20	US Douglas A-20 twin-engine bomber ("Havoc")
A-26	US Douglas A-26 twin-engine bomber ("Invader")
B-17	US Boeing B-17 four-engine bomber ("Flying Fortress")
B-24	US Consolidated B-24 four-engine bomber ("Liberator")
B-25	US North American B-25 twin-engine bomber ("Mitchell")
B-26	US Martin B-26 twin-engine bomber ("Marauder")

C-47	US Douglas C-47 twin-engine transport plane ("Skytrain")
*F-5	Photoreconnaissance version of P-38
P-38	US Lockheed P-38 twin-engine fighter ("Lightning")
P-47	US Republic P-47 single-engine fighter-bomber ("Thunderbolt")
P-51	US North American P-51 single-engine fighter ("Mustang")
P-61	US Northrop P-61 twin-engine night-fighter ("Black Widow")

British Designations

INFANTRY WEAPONS

Bren MG	British Bren .303 caliber machine gun (air-cooled/magazine-fed)
Vickers MG	British Vickers .303 caliber machine gun (water-cooled/belt-fed)
Mtr 2"	British 2" light mortar
Mtr 3"	British 3" medium mortar
*Mtr 4.2"	British 4.2" mortar
PIAT	British man-portable antitank grenade projector (Projectile, Infantry, Anti-Tank)

ANTIAIRCRAFT ARTILLERY

20mm AA	British 20mm towed antiaircraft gun
40mm AA (T)	British 40mm Bofors towed antiaircraft gun
40mm AA (SP) (A-15)	British 40mm Bofors self-propelled antiaircraft gun (Crusader tank chassis)
AA tank (20mm)	British twin 20mm Oerlikon self-propelled antiaircraft guns (Crusader tank chassis)
3.7" AA (T)	British 3.7" towed antiaircraft gun

ANTITANK (TOWED)

6-pr AT (T)	British 6-pounder (57mm) towed antitank gun
17-pr AT (T)	British 17-pounder (76.2mm) towed antitank gun

ARTILLERY

25-pr (T)	British 25-pounder (3.45") towed gun-howitzer
25-pr (SP)	British 25-pounder (3.45") self-propelled gun-howitzer ("Sexton")
5.5" Med Gun	British 5.5" towed gun-howitzer
7.2" How	British 7.2" towed howitzer

ANTITANK AND INFANTRY SUPPORT (SELF-PROPELLED)

17-pr AT (SP)	British 17-pounder (76.2mm) self-propelled antitank gun (Valentine tank or US M-10 chassis)

A-30 Challenger	British 17-pounder (76.2mm) self-propelled antitank gun (Cromwell tank chassis)

TANKS

Churchill 6-pr (A22)	British Churchill infantry tank w/6-pounder gun
Churchill 75mm (A22)	British Churchill infantry tank w/75mm gun
Churchill 95mm (A22)	British Churchill infantry tank w/95mm howitzer
Crocodile	British Churchill tank w/flamethrower and 75mm gun
Cromwell 75mm (A27)	British Cromwell cruiser tank w/75mm gun
Cromwell 95mm (A27)	British Cromwell cruiser tank w/95mm howitzer
Sherman 17-pr (M-4A4)	British M-4 cruiser tank w/17-pounder gun ("Firefly")

ARMORED CARS

BSA Daimler SC	British BSA Daimler scout car w/.303 caliber machine gun
Humber SC	British Humber scout car w/.303 caliber machine gun
Lt Recon Car	British Morris reconnaissance car w/.303 caliber machine gun
Daimler AC	British Daimler armored car w/2-pounder gun
T17E1 Staghound	British (US manufacture) armored car w/37mm gun
AEC AC	British AEC armored car w/75mm gun (Mk III) or w/6-pounder gun (Mk II)

SOFTSKIN VEHICLES

15 CWT trk	British 15 hundred weight truck
*15 CWT Water trk	British 15 hundred weight water truck
*15 CWT Water trl	British 15 hundred weight water trailer
*15 CWT trl	British 15 hundred weight trailer
*15 CWT Ambl	British 15 hundred weight ambulance
3 ton lorry	British 3 ton truck
*3 ton Dump	British 3 ton dump truck
Tractor C8	British Morris C8 wheeled artillery tractor
6 ton Scammel	British 6 ton wheeled prime mover Tractor

MISCELLANEOUS ARMORED VEHICLES

*Arm CV	British armored command/observation vehicle (US M-4 tank chassis)
T16	British Universal (Bren Gun) tracked carrier
ARV	British armored recovery vehicle (Churchill tank chassis)
*OP Tank	British armored command/observation vehicle (US M-4 tank chassis)
RAM K	British armored personnel carrier (Canadian
Ram	tank chassis)
*Avlb	British armored bridge-laying vehicle (Churchill tank chassis)

AIRCRAFT

Boston	British designation for US A-20
Halifax	British four-engine bomber
Lancaster	British four-engine bomber
Mitchell	British designation for US B-25
Mosquito	British twin-engine fighter-bomber/bomber/ photoreconnaissance aircraft/night-fighter
Mustang	British designation for US P-51
Spitfire	British single-engine fighter
Tempest	British single-engine fighter-bomber
Typhoon	British single-engine fighter-bomber

German Designations

INFANTRY WEAPONS

LMG	German MG-34/42 7.92mm light machine gun
HMG	German MG-34/42 7.92mm heavy machine gun
MMTR	German GrW 34 80mm mortar
HMTR	German GrW 42 120mm mortar
PF	German Panzerfaust 150-200mm man-portable antitank rocket launcher
PS	German Panzerschreck 88mm man-portable antitank rocket launcher (German equivalent of M-9 2.36" Rkt)
FLAM	German flamethrower

ANTIAIRCRAFT ARTILLERY

FLAK15(d) SP	German MK 151 15mm self-propelled antiaircraft gun (three barrel)
FLAK20	German Flak 38 20mm towed antiaircraft gun (single barrel)
FLAK20 SP	German Flak 38 20mm self-propelled antiaircraft gun (single barrel)
FLAK20(v)	German 20mm towed antiaircraft gun (four barrel)
FLAK20(v) SP	German 20mm self-propelled antiaircraft gun (four barrel)
FLAK37	German Flak 43 37mm towed antiaircraft gun
FLAK37 SP	German Flak 43 37mm self-propelled antiaircraft gun
FLAK37(z)	German Flak 43 37mm towed antiaircraft gun (two barrel)
FLAK88	German Flak 41 88mm towed antiaircraft gun
FLAK105	German Flak 38 105mm towed antiaircraft gun

ANTITANK (TOWED)

PAK50	German Pak 38 50mm towed antitank gun
PAK75	German Pak 40 75mm towed antitank gun
PAK88	German Pak 43/41 88mm towed antitank gun

ARTILLERY

LIG	German LeIG 18 & 37 75mm towed light infantry guns
HIG	German SIG 33 150mm towed heavy infantry gun

HIG SP	German SIG 33 150mm self-propelled heavy infantry gun
K75	German 75mm towed gun (various models -- FK 40; FK 16 n.a.; Pak 97/38; FK(r); FK 240(d))
PAK43ARTY	German Pak 43 88mm towed antitank gun (used as artillery)
K105	German Kan 18 105mm towed gun
K128	German Kan 44 128mm towed gun
K152	German 152mm towed gun-howitzer (various models -- KH 433(r) and K 425(f))
K170	German 170mm towed gun on howitzer carriage (various models -- Kan.i.Mrs.L.; Kan(t); K 3;
Kan	52)
LFH105	German 105mm towed howitzer (various models -- LeFH 18/40; LeFH 315(i))
LFH105 SP	German 105mm self-propelled howitzer
HFH150	German SFH 18 150mm towed howitzer
HFH150 SP	German SFH 18 150mm self-propelled howitzer
HFH122	German SFH 396(r) 122mm towed howitzer
HOW210	German Mrs 18 & 531(f) 210mm towed howitzers
HOW355	German 355mm towed howitzer (various models -- Mrs 1; Mrs 16(t); Mrs "Karl;" Mrs 601(f))
K240	German 240mm railway gun (various models -- Kan.Th.K; Kan.Th.Br.)
K280/5	German 280mm railway gun (various models -- K.5; K.5 gl; Kan.(f))
NW41	German Nebelwerfer 41 150mm towed six barrel rocket launcher
NW41 SP	German Nebelwerfer 41 150mm self-propelled ten barrel rocket launcher (Panzerwerfer 42)
NW42	German Nebelwerfer 42 210mm towed five barrel rocket launcher
NW42HVY	German Nebelwerfer 42 280mm (high explosive)/320mm (incendiary) towed six barrel rocket launcher (heavy)

(Note: (r) indicates weapon of Russian manufacture, (d) of Danish manufacture, (f) of French manufacture, (t) of Czechoslovakian manufacture, and (i) of Italian manufacture.)

ANTITANK AND INFANTRY SUPPORT (SELF-PROPELLED)

75HT	German half-track w/75mm howitzer (SdKfz 250/8 or 251/9)
PZJG-IV	German self-propelled antitank gun on PZ-IV chassis w/75mm L70 gun
PZJG-V	German self-propelled antitank gun on PZ-V chassis w/88mm L71 gun (Jagdpanther)
PZJG-38t	German self-propelled antitank gun on Czech
38t	chassis w/75mm L48 gun (Hetzer)

STG-III	German self-propelled assault gun w/75mm gun (various models -- Stg. III; Stg. IV; Pz Jg IV(k); PAK 39(SP))
BRUM	German Brummbaer self-propelled armored 150mm heavy infantry gun
STG105	German self-propelled assault gun w/105mm howitzer
STRM TIGER	German Sturm Moerser self-propelled assault vehicle on PZ-VI Tiger chassis w/380mm rocket launcher

TANKS

PZ-IV	German Mark IV medium tank -- all types (w/75mm gun)
PZ-V	German Mark V medium tank (Panther) -- all types (w/75mm gun)
PZ-VI	German Mark VIe heavy tank (Tiger I) -- all types (w/88mm gun)
PZ-VIb	German Mark VIb heavy tank (Tiger II) -- all types (w/88 mm gun)

ARMORED CARS

LAC	German armored cars w/20mm gun (all types -- includes SdKfz 250/9 half-track)
HAC	German armored cars w/50mm or 75mm gun (all types)

HALF-TRACKS

LHT	German SdKfz 250 half-track -- all types
MHT	German SdKfz 251 half-track -- all types

SOFTSKIN VEHICLES

KK	German motorcycle and motorcycle half-track
PKW	German wheeled personnel carrier
LKW	German wheeled cargo carrier
trl	German trailer
MAUL	German Maultier half-track cargo carrier
RCV	German SdKfz 100 & 9/1 recovery vehicle
1-5t PM	German 1-5 ton half-track prime movers
8-18t PM	German 8-18 ton half-track prime movers

RSO	German Raupenschlepper Ost and Schwere Wehrmacht Schlepper full-track prime movers
HDR	German horse-drawn vehicle

MISCELLANEOUS ARMORED VEHICLES

BGPZ-III	German armored recovery vehicles on PZ-III chassis
BGPZ-V	German armored recovery vehicle on PZ-V chassis
BEPZ-III/IV	German armored command/observation vehicles on PZ-III/IV chassis
BEPZ-V	German armored command/observation vehicle on PZ-V chassis
GOL	German Goliath wire-controlled demolition vehicle FKL German Funklenk armored radio-controlled demolition vehicle

AIRCRAFT

Ar-234	German Arado 234 twin-engine jet bomber
Fw-190	German Focke-Wulf 190 single-engine fighter-bomber
Fw-190D	German Focke-Wulf 190D single-engine fighter-bomber
Ju-52	German Junkers 52 twin-engine transport plane
Ju-88	German Junkers 88 twin-engine bomber
Me-109	German Messerschmitt 109 single-engine fighter-bomber
Me-262	German Messerschmitt 262 twin-engine jet fighter-bomber

An asterisk (*) indicates that 1) sufficient data was not available for this system to create a weapons record, 2) that the system was very similar to other systems for which data was recorded in a weapons record (this data should be used as a substitute), or 3) that the system appears in insignificant numbers in the ACSDB and does not warrant a weapons record.

The ACSDB Reference Data Base

The Ardennes Campaign Simulation Data Base (ACSDB) Reference Data Base is used to reference data recorded in the Air, Unit Location, Table of Organization and Equipment (T/O&E), and Weapons Data Bases. The Reference Data Base was originally intended to record sources used for all ACSDB data bases (the four listed above, plus the Unit and Unit Inventory Data Bases), but it proved impractical to use this data referencing system for the Unit and Unit Inventory Data Bases. Instead, written descriptions of the derivation of the Unit and Unit Inventory Data Bases are provided in various narratives (for example, see the ACSDB Unit Data Base and the ACSDB Unit Inventory Data Base narratives), and the Reference Data Base is used primarily for the four former data bases mentioned above.

Definitions of the fields in the ACSDB Reference Data Base are as follow:

- "Source Number" (No field number): Recorded under "Source Number" is a unique alphanumeric entry that matches the entry under "Sources" in the corresponding Air, Unit Location, T/O&E, and Weapons Data Base record.

  - Air reference records begin with "AA," "AB," or "AG," followed by three arabic numerals.

  - Unit Location reference records begin with "L," followed by a second letter or an arabic numeral, which is in turn followed by a series of three or four letters or arabic numerals, depending on the nationality and designation of the unit. The alphanumeric entries used for all Unit Location reference records are listed in the ACSDB Unit Location Data Base narrative.

  - T/O&E reference records begin with "TU," "TB," or "TG," followed by three arabic numerals.

  - Weapons reference records begin with "WA," "WB," or "WG," followed by three arabic numerals.

- "DATA BASE ITEMS" (Fields 1-20): Recorded under "DATA BASE ITEMS" are the data field(s) in the corresponding Air, Unit Location, T/O&E, or Weapons Data Base record for which data is referenced by the source listed under "BIBLIOGRAPHY NAMES."

- "BIBLIOGRAPHY NAMES" (Fields 1-20): Recorded under "BIBLIOGRAPHY NAMES" are the identifications of sources used for data listed under "DATA BASE ITEMS." All identifications entered under "BIBLIOGRAPHY NAMES" match exactly entries under

"Bibliography Name" in the Bibliography Data Base. The identifications are usually in abbreviated format, using author's names, archives identification numbers, microfilm numbers, etc.

- "PAGES" (Fields 1-20): Entered under "PAGES" are the relevant page number(s) of the sources listed under "BIBLIOGRAPHY" names.

- "Remarks" (No field number): Entered under "Remarks" is information which amplifies any other entries in the record and, in some cases, data in the corresponding Air, Unit Location, T/O&E, and Weapons Data Base record.

The ACSDB Bibliography Data Base

The Ardennes Campaign Simulation Data Base (ACSDB) Bibliography Data Base is used to record full bibliographic information on primary and secondary sources used for data in the various data bases of the ACSDB. The entries in the Bibliography Data Base describe sources referenced in both the Reference Data Base and in the various narratives used to describe the derivation of data (for example, see the ACSDB Unit Data Base and the ACSDB Unit Inventory Data Base narratives).

Note that some sources referenced in the narratives are not listed in the Bibliography Data Base. Instead, their full bibliographic references are contained in the narratives themselves. Thus, any document, record, book, or other source used to generate data for any part of the ACSDB is always fully referenced. In general, the numerous primary source records used for the generation of US Army personnel, equipment, and logistics data are referenced only in the narratives. These records are from the Washington Federal Records Center annex of the US National Archives, in Suitland, Maryland. Most other sources in the narratives are fully referenced in the Bibliography Data Base, and some sources are referenced in both the narratives and the Bibliography Data Base. Note that if sources mentioned in narratives are annotated only in the Bibliography Data Base, this fact is always highlighted at the beginning of the narrative.

The variety and great number of sources used in the ACSDB resulted in the use of both formats for full bibliographic annotations. For convenience and to minimize crossreferencing from a hard-copy to computer format, it was decided to use just the hard-copy format in some cases, especially those involving long lists of records from the same provenance. Nevertheless, full referencing of all sources used for data generation in the ACSDB is maintained.

Definitions of the Bibliography Data Base records are as follow.

- "Bibliography Name" ((Field 1): Recorded under "Bibliography Name" is a unique entry in abbreviated format, most commonly an author's name, an archives identification number, or a microfilm number, which identifies the source described under "Bibliography" (Field 2). An entry under "Bibliography Name" matches exactly an entry under "BIBLIOGRAPHY NAME" in the Reference Data Base records, or the abbreviation used for a source in a data estimation and referencing narrative.

- "Bibliography" (Field 2): Described under "Bibliography" is the full bibliographic annotation of the source. For

secondary sources, this includes author's name, title of the book, place and date of publishing, and publisher. For primary source records, this includes the title of the document and, in some cases, an archives or mircrofilm identification number. Generally, British records from the Public Records Office have their archives identification number recorded under "Bibliography." On the other hand, German sources from the German Archives in Freiburg, West Germany, or on microfilm at the US National Archives have their archives or microfilm number listed under "Record, Box, and File" (Field 3). Note that for all primary source records, archives or microfilm identification numbers are provided in either Field 2 or Field 3.

- "Record, Box, and File" (Field 3): Most commonly recorded under "Record, Box, and File" is the archival facility from which a primary source record was obtained. Typical entries are:

-- US Nat'l [National] Archives -- followed by a microfilm #
-- German Archives -- followed by a source =
-- British Public Records Office
-- Bolling AFB (Air Force Base) -- followed by a microfilm #
-- German Post-War Manuscript

The documents from the Bolling Air Force Base were obtained from microfilm records maintained at the Office of Air Force History at Bolling AFB in Washington, D.C. These records are mainly World War II records of the US Army Air Forces used for generation of data in Air Data Base records. The German Post-War Manuscripts are frequently-referenced sources for German unit location information in the Unit Location Data Base. These are kept on file at the US National Archives in Washington, D.C., and at the US Army Military History Institute library in Carlisle, Pennsylvania.

- "Remarks" (Field 4): Additional information on a bibliographic source may be recorded under "Remarks." This information includes comments on accuracy or importance of data found in a source. Due to space limitations in the computer format, it was not possible to record all relevant bibliographic information for some sources in Fields 1-3. In these cases, the additional information was recorded under "Remarks."

Sources Not Recorded in the Bibliography Data Base.

For bibliographic items not recorded in the Bibliography Data Base (most commonly US Army records, as discussed above), the same conventions used in the Bibliographic Data Base are used in the various data estimation and referencing narratives. Secondary sources are described by their author's name, title of the book, place and date of publishing, and publisher. Primary

sources are identified by all relevant archives identification. For US Army records from the Washington Federal Records Center, this includes record group (most commonly RG 407), box number, and the code used in the National Archives filing system, i.e., 304-1.1 (4th Armored Division, G-1 (Personnel) records.

ACSDB Time Conventions

As originally tasked in the Statement of Work of the Ardennes Campaign Simulation Data Base (ACSDB) Request for Proposal (RFP No. MDA903-87-R-0091), data was to be recorded on "the initial situation of each unit and subsequently a sequence of events by 24-hour intervals (0600-0559) throughout the campaign..." (see page 1 of the statement of work). This system of recording time was maintained to the extent possible in the compilation of the ACSDB, but, due to the nature of the primary source records used for data compilation, it was in some cases modified. This paper explains the time conventions in effect for personnel, equipment, and other data of the three nations in the ACSDB (US, British, and German data).

The primary reason for deviation from the required system of recording time was the nature of the records of the US Army units which participated in the Ardennes Campaign. In general, US Army units reported status of personnel, equipment, etc., at the end of the day, i.e., 1600 hours or 2400 hours. One of the main sources used for US personnel data, the 12th Army Group G-1 Daily Summary (US National Archives Record Group 407, Boxes 1753 and 1754), recorded personnel strengths of US divisions and non-divisional units as of 2400 hours of the reporting day. Therefore, in order to facilitate use of this important document for generating US personnel data, it was decided to change the time of the US personnel strengths in the ACSDB from as of beginning of the day (0600 hours) to as of the end of the day (0559 hours). In other words, the personnel strength of the 101st AbnD recorded in the ACSDB on 24 December 1944 is as of 0559 hours on 25 December, not 0600 hours on 24 December 1944. (To maintain consistency in the 24-hour time intervals, it was assumed that strengths as of 2400 hours in the above example would approximate the strengths as of 0559 hours on the next morning.)

Thus, the 24-hour time intervals for all data in the ACSDB are consistent (i.e., 0600 hours on one day to 0559 hours on the subsequent day). Casualties or equipment losses, personnel reinforcements or equipment replacements always occur in that 24-hour time interval. Only the time of the "initial situation," or status, of personnel, equipment, and logistics varies. These are explained below.

US Data.

- All personnel strengths, to include personnel by military occupational specialty (MOS) category, are as of 0559 hours of the date immediately subsequent to the indicated date. (The indicated date is the date recorded under "Date" in the Unit Data Base record.)

- All casualties and reinforcements (replacements and returns to duty) occur in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All medical facilities data (number of available and filled hospital beds) is as of 0559 hours of the date immediately subsequent to the indicated date. All "dynamic" medical facilities data (number of wounded personnel entering, number of disease and non-battle injury personnel entering, number of personnel died in, and number of personnel evacuated from hospitals) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All logistics data (ammunition, fuel, and other supply amounts on-hand and wet and dry transportation capacity) is as of 0559 hours of the date immediately subsequent to the indicated date. All "dynamic" logistics data (supplies consumed and received) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- The order of battle changes are as of 0600 hours of the indicated date.

- All equipment data (equipment on-hand and in-repair) is as of 0559 hours of the date immediately subsequent to the indicated date. All "dynamic" equipment data (equipment losses and gains) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All location data (headquarters location, front-line trace, location of other significant elements) is as of 0600 hours of the indicated date. All "dynamic" location data (displacement, distance opposed advanced, activities, and operations) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All air data occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

British Data.

- All personnel strengths are as of 0559 hours of the date immediately subsequent to the indicated date.

- All casualties and reinforcements (replacements and returns to duty) occur in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All logistics data (ammunition, fuel, and other supply amounts on-hand and wet and dry transportation capacity) is as of 0559 hours of the date immediately subsequent to the indicated date. All "dynamic" logistics data (supplies consumed and received) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- The order of battle changes are as of 0600 hours of the indicated date.

- All equipment data (equipment on-hand and in-repair) is as of 0600 hours of the indicated date. All "dynamic" equipment data (equipment losses and gains) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All location data (headquarters location, front-line trace, location of other significant elements) is as of 0600 hours of the indicated date. All "dynamic" location data (displacement, distance opposed advanced, activities, and operations) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All air data occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

German Data.

- All personnel strengths, to include personnel by military occupational specialty (MOS) category, are as of 0600 hours of the indicated date.

- All casualties and reinforcements (replacements and returns to duty) occur in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All logistics data (ammunition, fuel, and other supply amounts on-hand and wet and dry transportation capacity) is as of 0600 hours of the indicated date. All "dynamic" logistics data (supplies consumed and received) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- The order of battle changes are as of 0600 hours of the indicated date.

- All equipment data (equipment on-hand and in-repair) is as of 0600 hours of the indicated date. All "dynamic" equipment data (equipment losses and gains) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All location data (headquarters location, front-line trace, location of other significant elements) is as of 0600 hours of the indicated date. All "dynamic" location data (displacement, distance opposed advanced, activities, and operations) occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.

- All air data occurs in the 24 hours between 0600 hours of the indicated date and 0559 hours of the immediately subsequent date.